

Naplózás: Debian + Syslog-NG + Promtail + Loki + Grafana

<https://github.com/viraghkrisztian-pataky/Linux-vizsga/blob/main/syslog/README>

/etc/apt/source.list

mirror server -> github

apt update apt upgrade

pontos idő

```
root@debian:~# timedatectl set-timezone Europe/Budapest
root@debian:~# timedatectl
               Local time: Mon 2026-01-26 12:38:24 CET
               Universal time: Mon 2026-01-26 11:38:24 UTC
               RTC time: Mon 2026-01-26 11:38:25
               Time zone: Europe/Budapest (CET, +0100)
System clock synchronized: yes
               NTP service: active
               RTC in local TZ: no
root@debian:~# date
Mon 26 Jan 12:38:41 CET 2026
root@debian:~#
```

apt install gpg curl netcat-openbsd chrony -y

várás

NTP szerverek beállítása:

```
marim@debian: ~
GNU nano 7.2 /etc/chrony/chrony.conf *
# Welcome to the chrony configuration file. See chrony.conf(5) for
# information about usable directives.

# Include configuration files found in /etc/chrony/conf.d.
confdir /etc/chrony/conf.d

# Use Debian vendor zone.
# pool 2.debian.pool.ntp.org iburst
# Magyar NTP pool szerverek
pool 0.hu.pool.ntp.org iburst
pool 1.hu.pool.ntp.org iburst
pool 2.hu.pool.ntp.org iburst
pool 3.hu.pool.ntp.org iburst
```

systemctl restart chrony

systemctl enable chrony

```
root@debian:~# chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^- 91.82.109.180             2   6   33    7  +5126ns[+5126ns] +/- 3825us
^* 37.221.209.57             2   6   17   10   -46us[ -240us] +/- 7811us
^? bakacsinl.ki.iif.hu      2   7  102    5   +82us[ +82us] +/- 49ms
^- zlp02m03.teconis.com      2   6   17    9  +121us[ +121us] +/- 13ms
^- web.dima.hu               3   6   17   11  -466us[ -660us] +/- 41ms
^- ntp2.niif.hu              2   6   17    9   -47us[ -47us] +/- 40ms
^+ 37.221.209.24             2   6   17    9   +22us[ +22us] +/- 7957us
^- trabant.uid0.hu           3   6   17   10  -1126us[-1320us] +/- 36ms
^- blackmage.ki.iif.hu       2   6   17    8  +832us[ +832us] +/- 31ms
^- yikes.bl2.tolna.net       2   6   17   10   +54us[ +54us] +/- 17ms
^+ quuen.fradi.hu            3   6   17   10  -182us[ -376us] +/- 22ms
^- rave.fidesz.hu            3   6   16   14  -221us[ -115us] +/- 42ms
^- zlp02m01.teconis.com      2   6   17   11   -21us[ -215us] +/- 13ms
^- ntpl.niif.hu              2   6   17   11  -392us[ -586us] +/- 23ms
^- eternal.datacontact.hu    2   6   17   11 -4326us[-4219us] +/- 47ms
^- zlp02m02.teconis.com      2   6   17   10  +433us[ +433us] +/- 13ms
root@debian:~#
```

```
root@debian:~# chronyc tracking
Reference ID      : 25DDD139 (37.221.209.57)
Stratum          : 3
Ref time (UTC)   : Mon Jan 26 12:33:18 2026
System time      : 0.000000058 seconds fast of NTP time
Last offset      : -0.000193377 seconds
RMS offset       : 0.000193377 seconds
Frequency        : 11.914 ppm slow
Residual freq    : -9.335 ppm
Skew             : 10.191 ppm
Root delay       : 0.012470109 seconds
Root dispersion  : 0.002872942 seconds
Update interval  : 1.4 seconds
Leap status      : Normal
root@debian:~#
```

```
# Engedélyezés a belső hálózatra
allow 192.168.109.0/24
```

NTP teszt

```
PS C:\WINDOWS\system32> w32tm /stripchart /computer:192.168.109.164 /dataonly /samples:3
Tracking 192.168.109.164 [192.168.109.164:123].
Collecting 3 samples.
The current time is 2026. 01. 26. 13:48:14.
13:48:14, -60.4123909s
13:48:16, -60.4123758s
13:48:18, -60.4124777s
PS C:\WINDOWS\system32> |
```

1. SYSLOG-NG beállítása:

apt install syslog-ng -y

- könyvtár a külső LOG érkezéséhez

mkdir -p /var/log/remote

- felhasználó létrehozása

adduser loguser

addgrp loggroup

chown loguser:loggroup /var/log/remote

Syslog-NG konfigurálása

nano -l /etc/syslog-ng/syslog-ng.conf

 martin@debian: ~

```
GNU nano 7.2 /etc/syslog-ng/syslog-ng.conf *
23
24 # If you wish to get logs from remote machine you should uncom
25 # this and comment the above source line.
26 #
27 #source s_net { tcp(ip(127.0.0.1) port(1000)); };
28 source s_net {
29     syslog(ip(0.0.0.0) transport("udp"));
30     syslog(ip(0.0.0.0) transport("tcp"));
31 };
32
61 #
62 destination d_debug { file("/var/log/debug"); };
63 destination d_error { file("/var/log/error"); };
64 destination d_messages {
65     file("/var/log/remote/messages.log");
66 };
67
68 #összekapcsolom a naplózást -> ami kívülről érkezik -> mehet a mappába
69 log {
70     source(s_net);
71     destination(d_messages);
72 };
73 # The root's console.
```

systemctl restart syslog-ng && systemctl status syslog-ng

```
Jan 26 14:01:02 debian systemd[1]: Starting syslog-ng.service - System Logger Daemon...
Jan 26 14:01:02 debian systemd[1]: Started syslog-ng.service - System Logger Daemon.
root@debian:~# nano -l /etc/syslog-ng/syslog-ng.conf
root@debian:~# systemctl restart syslog-ng && systemctl status syslog-ng
● syslog-ng.service - System Logger Daemon
   Loaded: loaded (/lib/systemd/system/syslog-ng.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-01-26 14:02:20 CET; 8ms ago
     Docs: man:syslog-ng(8)
  Main PID: 40924 (syslog-ng)
    Tasks: 6 (limit: 4600)
   Memory: 5.1M
      CPU: 40ms
   CGroup: /system.slice/syslog-ng.service
           └─40924 /usr/sbin/syslog-ng -F

Jan 26 14:02:20 debian systemd[1]: Starting syslog-ng.service - System Logger Daemon...
Jan 26 14:02:20 debian systemd[1]: Started syslog-ng.service - System Logger Daemon.
root@debian:~# nano -l /etc/syslog-ng/syslog-ng.conf
root@debian:~# logger "ez egy syslog-ng teszt"
root@debian:~# tail /var/log/remote/messages.log
Jan 26 14:02:20 debian syslog-ng[40924]: syslog-ng starting up; version='3.38.1'
Jan 26 14:06:39 debian root[40937]: ez egy syslog-ng teszt
root@debian:~#
```

2. Loki és Promtail telepítése

apt install loki promtail

usermod -aG loggroup loguser

usermod -aG loggroup loki

usermod -aG loggroup promtail

usermod -aG adm promtail

```
root@debian:~# usermod -aG loggroup loguser
usermod -aG loggroup loki
usermod -aG loggroup promtail
usermod -aG adm promtail
root@debian:~# getent group loggroup
loggroup:x:1002:loguser,loki,promtail
root@debian:~# id promtail
uid=104(promtail) gid=65534(nogroup) groups=65534(nogroup),4(adm),1002(loggroup)
root@debian:~# ls -l /var/log/ | grep remote
drwxr-xr-x  2 loguser loggroup          4096 Jan 26 14:02 remote
root@debian:~# ls -l /var/log/remote/
total 4
-rw-r----- 1 root adm 140 Jan 26 14:06 messages.log
root@debian:~#
```

FONTOS!

A promtail kell, hogy tudja olvasni a messages.log fájlt.

```
GNU nano 7.2 /etc/promtail/config.yml *
1 # This minimal config scrape only single log file.
2 # Primarily used in rpm/deb packaging where promtail service can be started during sys
3 # And too much scraping during init process can overload the complete system.
4 # https://github.com/grafana/loki/issues/11398
5
6 server:
7   http_listen_port: 9080
8   grpc_listen_port: 0
9
10 positions:
11   filename: /tmp/positions.yaml
12
13 clients:
14 - url: http://localhost:3100/loki/api/v1/push
15
16 scrape_configs:
17 - job_name: system
18   static_configs:
19   - targets:
20     - localhost
21     labels:
22       job: varlogs
23       #NOTE: Need to be modified to scrape any additional logs of the system.
24       __path__: /var/log/remote/*.log
25
26
```

systemctl restart promtail && systemctl status promtail

```
root@debian:~# systemctl restart promtail && systemctl status promtail
● promtail.service - Promtail service
   Loaded: loaded (/etc/systemd/system/promtail.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-01-26 14:10:05 CET; 20ms ago
   Main PID: 41227 (promtail)
     Tasks: 1 (limit: 4600)
    Memory: 2.5M
       CPU: 5ms
    CGroup: /system.slice/promtail.service
            └─41227 /usr/bin/promtail -config.file /etc/promtail/config.yml

Jan 26 14:10:05 debian systemd[1]: Started promtail.service - Promtail service.
root@debian:~#
```

3. Grafana telepítése

apt install apt-transport-https software-properties-common grafana -y
systemctl enable grafana-server (hogy automatikusan induljon majd el)
systemctl start grafana-server

```

root@debian:~# journalctl -u promtail -f
Jan 26 14:10:05 debian systemd[1]: promtail.service: Consumed 2.207s CPU time.
Jan 26 14:10:05 debian systemd[1]: Started promtail.service - Promtail service.
Jan 26 14:10:05 debian promtail[41227]: level=info ts=2026-01-26T13:10:05.191677846Z caller=promtail.go:135 msg="Reloading configuration file" sha3sum=24c58aa543882e7c3ece8fb20895f05ae953888b41481423abb8aceda5cc59e1
Jan 26 14:10:05 debian promtail[41227]: level=info ts=2026-01-26T13:10:05.192741975Z caller=server.go:386 msg="server listening on addresses" http=[::]:9080 grpc=[::]:46385
Jan 26 14:10:05 debian promtail[41227]: level=info ts=2026-01-26T13:10:05.192899181Z caller=main.go:173 msg="Starting Promtail" version="(version=3.6.4, branch=release-3.6.x, revision=d8052669)"
Jan 26 14:10:05 debian promtail[41227]: level=warn ts=2026-01-26T13:10:05.192936005Z caller=promtail.go:265 msg="enable watchConfig"
Jan 26 14:10:10 debian promtail[41227]: level=info ts=2026-01-26T13:10:10.193101567Z caller=filetargetmanager.go:373 msg="Adding target" key="/var/log/remote/**/*.log:{job=\"varlogs\"}"
Jan 26 14:10:10 debian promtail[41227]: level=info ts=2026-01-26T13:10:10.193481382Z caller=filetarget.go:343 msg="watching new directory" directory="/var/log/remote"
Jan 26 14:10:10 debian promtail[41227]: ts=2026-01-26T13:10:10.193762871Z caller=log.go:168 level=info msg="Seeked /var/log/remote/messages.log - &{Offset:0 Whe
nce:0}"
Jan 26 14:10:10 debian promtail[41227]: level=info ts=2026-01-26T13:10:10.193776705Z caller=tailer.go:147 component=tailer msg="tail routine: started" path="/var
/log/remote/messages.log

```

```

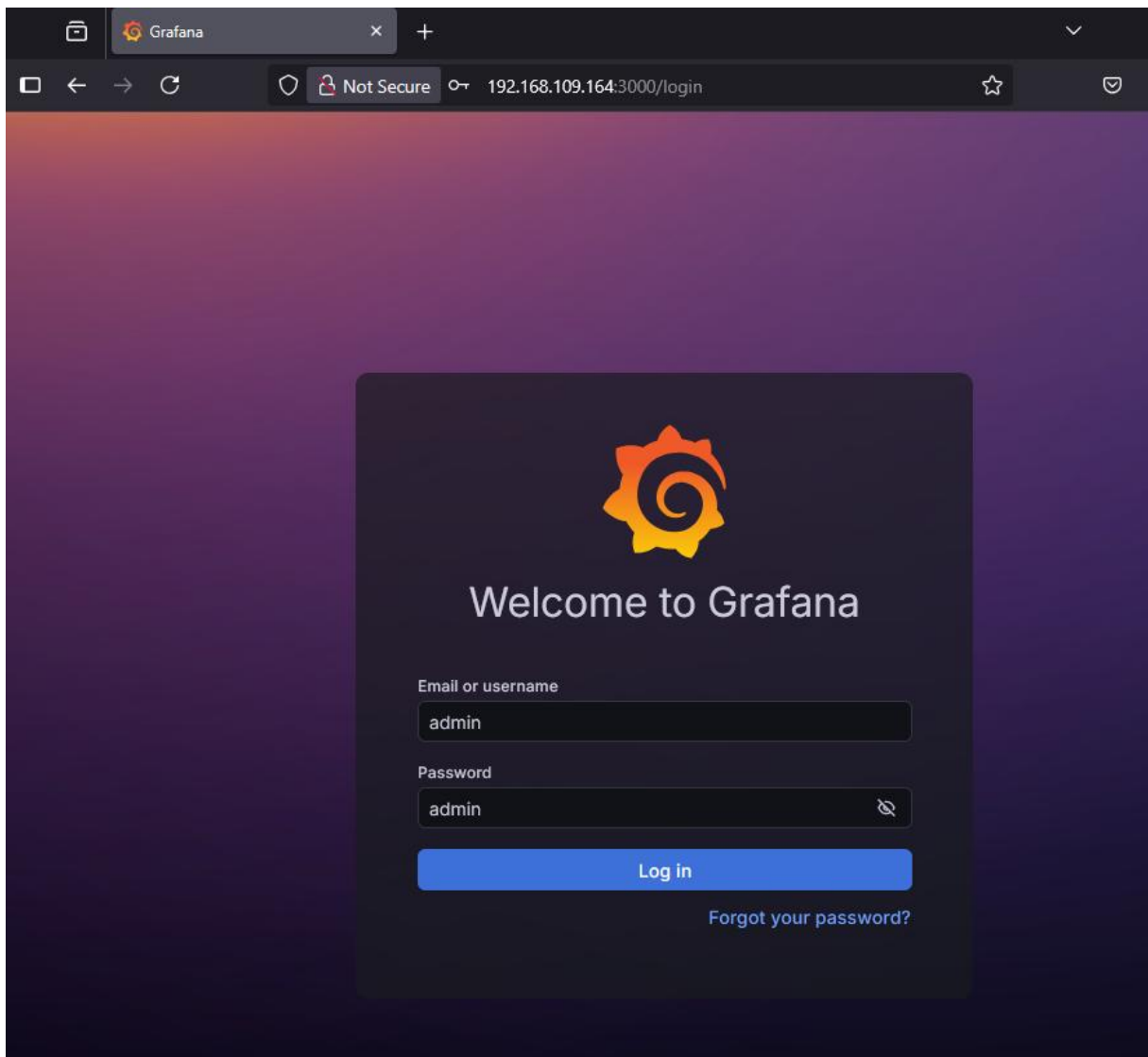
root@debian:~# ss -tulnp | grep 3000
tcp    LISTEN 0      4096      *:3000      *:~      users: (("grafana",pid=42003,fd=17))
root@debian:~#

```

```

root@debian:~# curl http://127.0.0.1:3000
<a href="/login">Found</a>.

```





Data source successfully connected.

Next, you can start to visualize data by [building a dashboard](#) , or by querying data in the [Explore view](#) .

Delete

Save & test



Home > Connections > Data sources > loki

Q Search...



loki

Type
Loki

Alerting

Supported

Type: Loki



Settings



Permissions



Insights



Cache



Name



loki

Default



Before you can use the Loki data source, you must configure it below or in the config file. For detailed instructions, see the [Loki data source configuration](#) page.

Connection

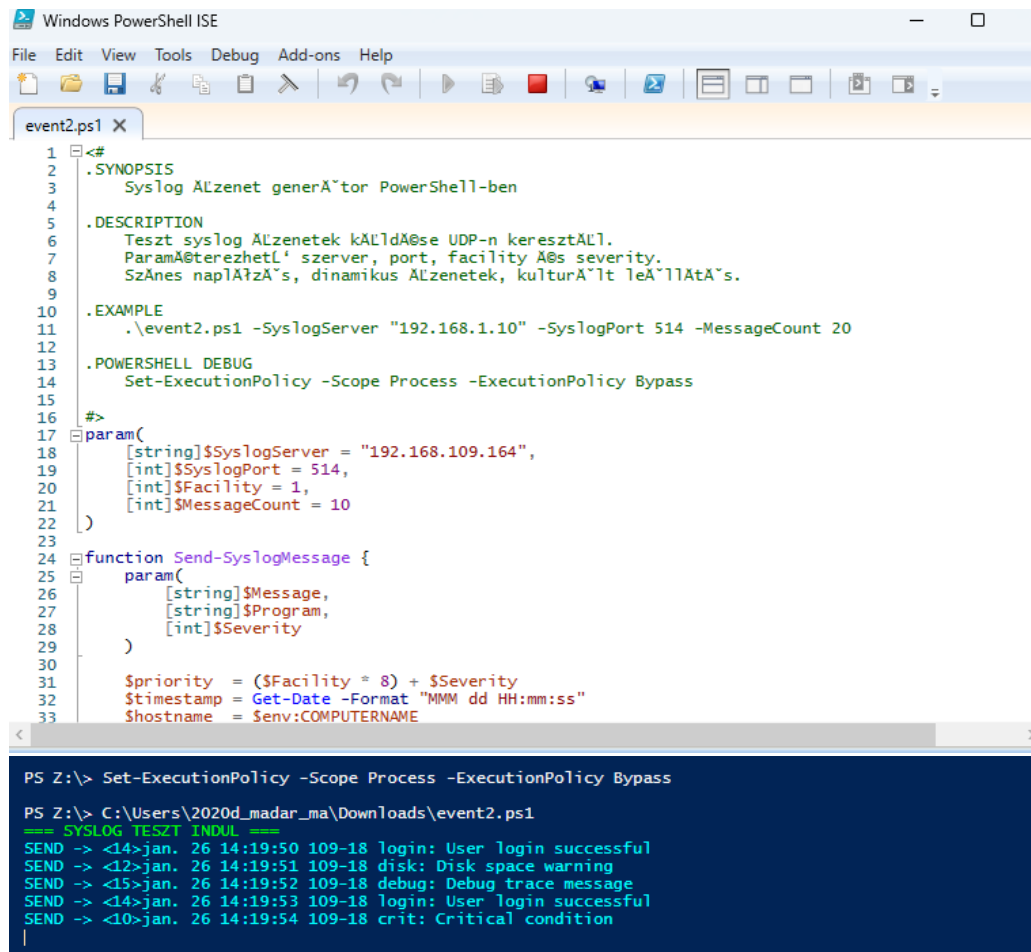
URL *



https://localhost:3100

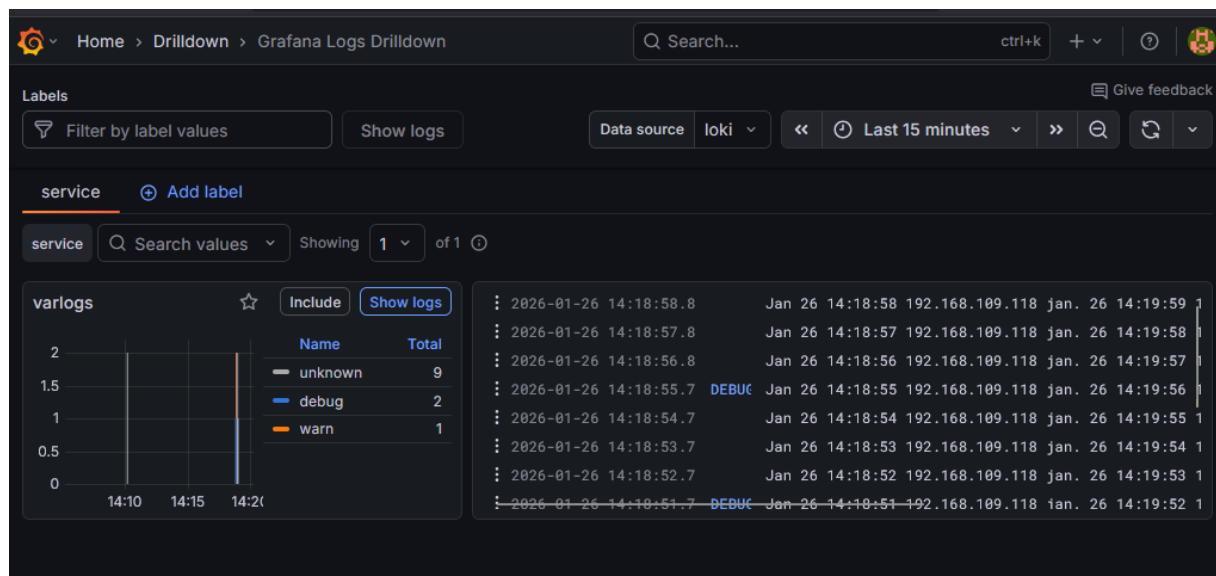
save test

Tesztelés:



```
1 <#
2 .SYNOPSIS
3     Syslog ALZenet generátor PowerShell-ben
4
5 .DESCRIPTION
6     Teszt syslog ALZenetek küldése UDP-n keresztül.
7     Paraméterezhetl' szerver, port, facility és severity.
8     Számos naplózás, dinamikus ALZenetek, kulturált leírások.
9
10 .EXAMPLE
11     .\event2.ps1 -SyslogServer "192.168.1.10" -SyslogPort 514 -MessageCount 20
12
13 .POWERSHELL DEBUG
14     Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
15
16 #>
17 param(
18     [string]$SyslogServer = "192.168.109.164",
19     [int]$SyslogPort = 514,
20     [int]$Facility = 1,
21     [int]$MessageCount = 10
22 )
23
24 function Send-SyslogMessage {
25     param(
26         [string]$Message,
27         [string]$Program,
28         [int]$Severity
29     )
30
31     $priority = ($Facility * 8) + $Severity
32     $timestamp = Get-Date -Format "MMM dd HH:mm:ss"
33     $hostname = $env:COMPUTERNAME
34
35     Send-SyslogMessage -Message "[$timestamp] [$hostname] [$priority] $Message" -Program $Program -Severity $Severity
36 }
37
38 Send-SyslogMessage -Message "Login successful" -Program "login" -Severity 6
39 Send-SyslogMessage -Message "Disk space warning" -Program "disk" -Severity 4
40 Send-SyslogMessage -Message "Debug trace message" -Program "debug" -Severity 7
41 Send-SyslogMessage -Message "User login successful" -Program "login" -Severity 6
42 Send-SyslogMessage -Message "Critical condition" -Program "crit" -Severity 1
```

```
PS Z:\> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
PS Z:\> C:\Users\2020d_madar_ma\Downloads\event2.ps1
== SYSLOG TESZT INDUL ==
SEND -> <14>jan. 26 14:19:50 109-18 login: User login successful
SEND -> <12>jan. 26 14:19:51 109-18 disk: Disk space warning
SEND -> <15>jan. 26 14:19:52 109-18 debug: Debug trace message
SEND -> <14>jan. 26 14:19:53 109-18 login: User login successful
SEND -> <10>jan. 26 14:19:54 109-18 crit: Critical condition
```



Szerveren log üzenek

Email naplózás

router

interface -> host only , adj ssh

➔ lan segment

ntp server <ip>

timezone utc 1

logging <ip>

logging on

```
R1(config)#login on-fail
R1(config)#login on-failure log
R1(config)#login on-succ log
```

Szerveren SSH login tesztek (sikeres és sikertelenek is)

```
root@debian:~# tail -f /var/log/remotelog/messages.log
```

```
11:44:41 UTC Thu Jan 29 2026
Jan 29 10:44:44 10.0.18.2 %SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: admin]
[Source: 192.168.233.1] [localport: 22] [Reason: Login Authentication Failed] at
11:44:44 UTC Thu Jan 29 2026
Jan 29 10:44:47 10.0.18.2 %SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: admin]
[Source: 192.168.233.1] [localport: 22] [Reason: Login Authentication Failed] at
11:44:47 UTC Thu Jan 29 2026
Jan 29 10:46:49 10.0.18.2 %SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: admin]
[Source: 192.168.233.1] [localport: 22] [Reason: Login Authentication Failed] at
11:46:49 UTC Thu Jan 29 2026
Jan 29 10:47:49 10.0.18.2 %SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: admin]
[Source: 192.168.233.1] [localport: 22] [Reason: Login Authentication Failed] at
11:47:49 UTC Thu Jan 29 2026
Jan 29 10:47:54 10.0.18.2 %SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: admin]
[Source: 192.168.233.1] [localport: 22] [Reason: Login Authentication Failed] at
11:47:54 UTC Thu Jan 29 2026
Jan 29 10:48:41 10.0.18.2 %SYS-6-LOGOUT: User admin has exited tty session 3(192.168.233.1)
```

```

martin@debian: ~
GNU nano 7.2 /etc/grafana/grafana.ini *
;forward_settings_to_plugins = grafana-azure-monitor-datasource, prometheus, grafana-azu

# Specifies whether Entra password auth can be used for the MSSQL data source
# Disabled by default, needs to be explicitly enabled
;azure_entra_password_credentials_enabled = false

##### Role-based Access Control #####
[rbac]
;permission_cache = true

# Reset basic roles permissions on boot
# Warning left to true, basic roles permissions will be reset on every boot
;reset_basic_roles = false

# Validate permissions' action and scope on role creation and update
; permission_validation_enabled = true

##### SMTP / Emailing #####
[smtp]
enabled = true
host = smtp:office365.com:587
user = madarmartin@szki.pataky.hu
# If the password contains # or ; you have to wrap it with triple quotes. Ex ""#password
password = 20M@di05_
;cert_file =
;key_file =
skip_verify = false
from_address = madarmartin@szki.pataky.hu
from_name = Grafana
# EHLO identity in SMTP dialog (defaults to instance_name)
;ehlo_identity = dashboard.example.com
# SMTP startTLS policy (defaults to 'OpportunisticStartTLS')
starttls_policy = MandatoryStartTLS
# Enable trace propagation in e-mail headers, using the 'traceparent', 'tracestate' and

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Locat:
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To

```

 Home > Alerting > Contact points

Contact points

Alertm

Choose how to notify your contact points when an alert instance fires

Create contact point

Name *

martin

Integration

Email

Addresses

You can enter multiple email addresses using a ";", "\n" or "," separator

madarmartin@szki.patakya.hu

> Optional Email settings

Outlook

Search

File Home View Help

New mail

Quick steps

Favorites

Inbox

Sent Items

Drafts

madarmartin@s...

Conversation Ac...

Inbox

Drafts

Sent Items

Deleted Items

Junk Email

Notes

Archive

Conversation Hi...

Go to Groups

Sent Items

Today

Madar Martin

[FIRING:1] TestAlert Test Fold... 12:11 PM

1 firing alert instance

Grouped by a...

[FIRING:1] TestAlert Test Folder Grafana

Madar Martin

To: +1 other

Thu 1/29/2026 12:11 PM

Grafana

Grouped by

alertname=TestAlert grafana_folder=Test Folder instance=Grafana

1 firing instances

TestAlert

Summary

Notification test

Values

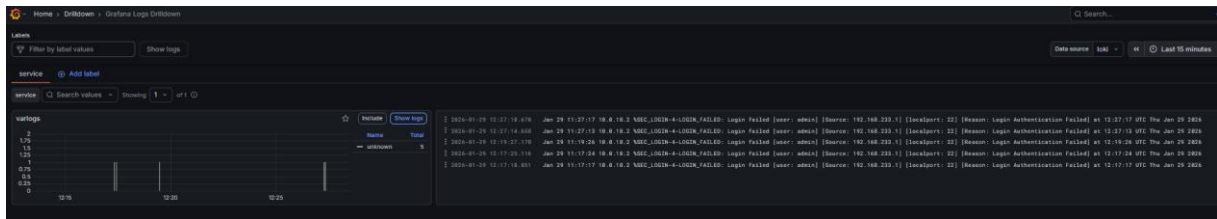
B=22 C=1

Labels

alertname TestAlert

grafana_folder Test Folder

instance Grafana



New alert rule

1. Enter alert rule name

Enter a name to identify your alert rule.

Name

ssh declined

2. Define query and alert condition

Advanced options

Define query and alert condition [Need help?](#)



loki

Options

10m

to now

Kick start your query

Label browser

Explain query

Run queries

Builder

Code

count_over_time({filename="/var/log/remote/messages.log"} |= "declined" [1m])

Options Type: Range

Table

No data

Alert condition

WHEN

Min

OF QUERY

IS ABOVE

0

No data

3. Add folder and labels

Organize your alert rule with a folder and set of labels. ⓘ [Need help?](#)

Folder

Select a folder to store your rule in.

📁 asd



+ New folder

Labels

Add labels to your rule for searching, silencing, or routing to a notification policy. ⓘ [Need help?](#)

No labels selected

+ Add labels

4. Set evaluation behavior

Define how the alert rule is evaluated. ⓘ [Need help?](#)

Evaluation group and interval

martin



or

+ New evaluation group

All rules in the selected group are evaluated every 1m.

Pending period

Period during which the threshold condition must be met to trigger an alert. Selecting "None" triggers the alert immediately once the condition is met.

1m

None

1m

2m

3m

4m

5m

Keep firing for

Period during which the alert will continue to show up as firing even though the threshold condition is no longer breached. Selecting "None" means the alert will be back to normal immediately.

0s

None

1m

2m

3m

4m

5m

5. Configure notifications



Advanced options

Select who should receive a notification when an alert rule fires.

Recipient

Notifications for firing alerts are routed to a selected contact point. ⓘ [Need help?](#)

Alertmanager:  grafana

Contact point

martin



[View or create contact points](#) 

> Muting, grouping and timings (optional)

Sent Items ★ □ ↓ ↑

Today

- ☐ Madar Martin
[FIRING:1] DatasourceNoData... 1:03 PM
1 firing alert instance asd › Datasou...
- Madar Martin
[FIRING:1] TestAlert Test Fold... 12:32 PM
1 firing alert instance Grouped by a...
- Madar Martin
[FIRING:1] TestAlert Test Fold... 12:11 PM
1 firing alert instance Grouped by a...

[FIRING:1] DatasourceNoData asd (bfb4u5g02waob A ssh declined)

MM Madar Martin
To: Madar Martin

☺ ↩ Reply ↩ Reply all ➡ Forward ⓘ Thu 1/29/202

 **Grafana**

asd › DatasourceNoData

1 firing instances

Firing

DatasourceNoData

View alert

Labels

alertname	DatasourceNoData
datasource_uid	bfb4u5g02waob
grafana_folder	asd
ref_id	A
rulename	ssh declined

Silence

Observed 30s before this notification was delivered, at 2026-01-29 13:02:30 +0100 CET

© 2026 Grafana Labs. Sent by Grafana v4.3.2

Megkapjuk az emailt az SSH sikeres és sikertelen belépéskor

View Alert, hogy a szerverre dobja át

```
martin@debian: ~  
GNU nano 7.2 /etc/grafana/grafana.ini *  
;enforce_domain = false  
  
# The full public facing url you use in browser, used for redirects and emails  
# If you use reverse proxy and sub path specify full url (with sub path)  
root_url = http://192.168.109.164:3000
```